

Performance Testing

Date	7 July 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional Performance Testing
Target Module / API	Credit Card Eligibility Prediction (/predict)
Test Environment	Local Flask Server / Render Deployment
Test Date	07 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid applicant information	1	30	Prediction generated successfully
2	Submit multiple prediction requests sequentially	10	60	Stable response without errors
3	Submit invalid or incomplete input	5	30	Input validation prevents incorrect requests
4	Continuous prediction requests	20	120	Application remains responsive with acceptable response time

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Average)	< 2 seconds	0.82 sec	Pass	Fast prediction response
2	Response Time (Maximum)	< 5 seconds	1.67 sec	Pass	Within acceptable limit
3	Throughput	> 10 Requests/sec	14 Requests/sec	Pass	Stable request handling
4	Error Rate	< 1%	0%	Pass	No failed prediction requests
5	CPU Utilization	< 80%	38%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	42%	Pass	Stable memory consumption

Step 4: Observations & Analysis

- The application responded consistently to prediction requests without noticeable delays.
- Input validation prevented invalid data from reaching the prediction engine.
- The Flask backend successfully handled multiple prediction requests with stable response times.
- CPU and memory usage remained within acceptable limits during testing.
- No prediction failures or application crashes were observed.
- The deployed application demonstrated reliable performance for the expected user workload.
- The application is suitable for small to medium-scale usage and can be further optimized for higher traffic if required.